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Voice-to-3D: Generative AI for Real-Time Generation of 3D Objects for Augmented Reality

INTRODUCTION

AR is only as rich as its 3D content

Every 3D asset an AR application displays must be modelled in advance, so the experience is limited to a fixed catalogue. Generative models now build textured 3D objects in seconds, but **no single model goes from speech to a manipulable 3D object**.

Fixed catalogue

Users only see what was pre-authored.

Slow & costly

Hand-modelling takes hours per object.

Caps creativity

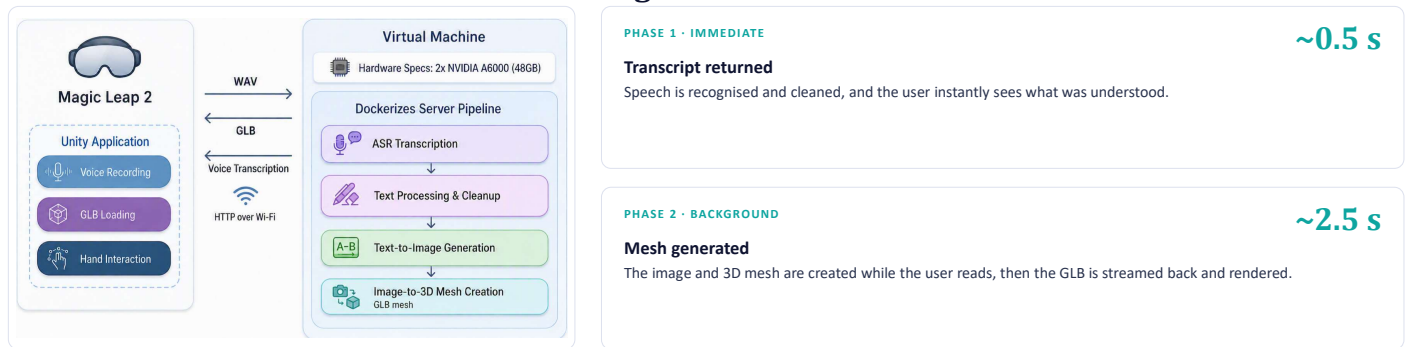
No open-ended, on-demand content.

THE QUESTION

What if a user could just say what they want, and watch it appear in 3D?

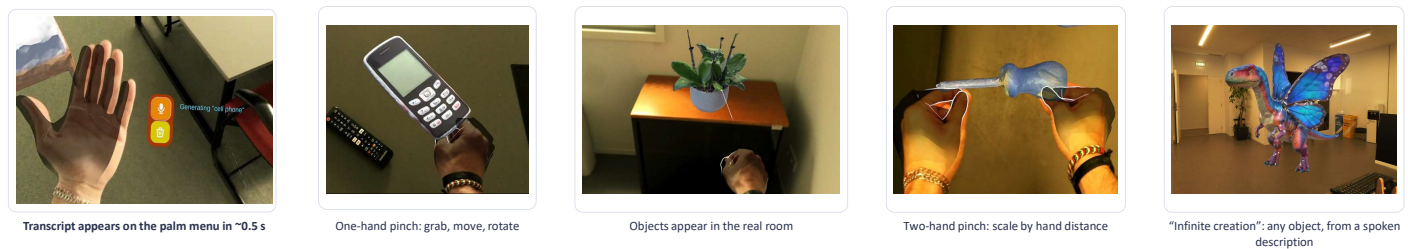
HOW IT WORKS

From voice to hologram in about three seconds



THE APPLICATION

Speak it, see it appear, grab it



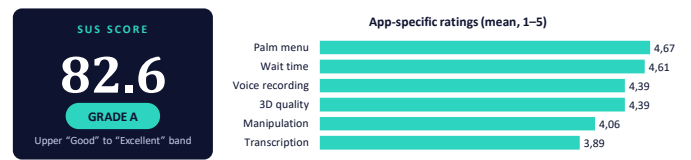
WHAT IT GENERATES

Spoken word, textured 3D asset



EVALUATION

Tested with 18 users, mostly novices



PRIOR WORK

Compared with prior 3D generative AI systems

System	Latency	Evaluation
Dream Mesh [1]	30–40 min	None
Transcending Dim. [2]	~68 s	SUS 69.6
Matrix [3]	< 50 s	SUS 69.6
Say It, See It [4]	18–74 s	Likert
This work	~3 s	SUS 82.6

References [1] Weng et al., AIXR 2024. [2] Behravan et al., HCII 2025. [3] Behravan & Gračanin, IEEE VRW 2025. [4] Xiu et al., ISMAR-Adjunct 2025.

CONCLUSION

Voice-driven 3D generation in AR is feasible today

Findings a mean SUS of 82.64 (grade A) with a mostly novice cohort; the palm menu and the generation wait time scored highest, confirming the speed-constrained choice of SF3D.

Limitations latency depends heavily on the GPU server; the benchmark used three objects and five prompts, without failure cases or statistical analysis; the application detects only English language.

Next steps native-language ASR, a user-selectable quality-speed mode, prompt conditioning, and finer per-object scene management.

Human Perception, Cognition & XR Methods

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